

Wenjie Luo

Agent Application Engineer | Graduating Jan 2027

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PROFILE

Build Python and TypeScript agent applications around deterministic tools and human approval. Translate workflows into APIs, state transitions, traces and automated acceptance checks.

EDUCATION

- National University of Singapore | MSc Civil Engineering**Aug 2025 - Jan 2027
- Transport focus; research in transportation information and control.
- Chang'an University | BEng Transportation Engineering**Sep 2021 - Jun 2025
- GPA 3.16/5.0; approximately top 15% of the major.

EXPERIENCE

- Far East Facade | Project Engineer, Engineering Department**Jun 2026 - Present
- Built a Python workflow to parse supplier shipment and test emails into arrival batches and inspection outcomes, supporting delivery plans for 15 material types.
- Caixun Technology | Product Intern**Jun 2025 - Aug 2025
- Wrote 3 PRDs with acceptance criteria and built 5 Axure prototypes adopted into development; followed functional checks and helped close more than 10 defects.

PROJECTS

- Civil Buddy | Agent Runtime and Tools**Jul 2026 - Present
- Implemented session scheduling, a tool engine, policy checks and recovery. Workflows cover 66 role SOPs, with human confirmation before high-risk writes and audit records for failures.
- Connected packing calculation, confirmation and visualization; all 128 deterministic evaluations passed. A separate shadow evaluation compares steps and tool-calling paths, including a keyless policy fallback.
- Glass Box | Agent Observability Middleware**Aug 23, 2026 - Present
- Built an observability layer on the official starter: parent-child span waterfalls expose tools, tokens, estimated costs and retries, with trace-query endpoints and OTLP/JSON export.
- Created keyless replay demonstrations and policy tests for 22 bypass attempts and 11 allowed cases. CI runs 150 cases across 21 files plus type checks, builds and smoke checks on Node 22/24.
- EDA Copilot | Traceable Engineering Q&A**Sep 2026
- Ran the OpenROAD nangate45/gcd example from RTL to GDSII and indexed reports, logs and workflow documentation into 1,215 chunks with line-level provenance.
- Implemented BM25, TF-IDF and hybrid retrieval with answers assembled from source excerpts; integrated retrieval and guard evaluations into CI to check definitions, run identity and evidence.

SKILLS

Development: Python, TypeScript, React, Fastify/FastAPI, SQL, Git, CI
Agents: tool calling, session scheduling, human approval, traces/OTLP, policy gates, recovery